

MBA RAPID FIRE

CAT 2025 - FINAL LAP

FOR MBA ASPIRANTS

Lecture - 02

Arithmetic - 2

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CAT 99.95 %ile



TOPICS TO BE COVERED

1

Important Concepts + Shortcuts

2

PRACTICE QUESTIONS

CHEAT SHEETS

P AND L





$$SP - CP = \text{Profit}$$

$$\text{Profit}\% = \frac{\text{Profit}}{CP} \times 100$$

$$CP \left[1 + \frac{\text{GAIN}\%}{100} \right] = SP$$



$$CP - SP = \text{loss}$$

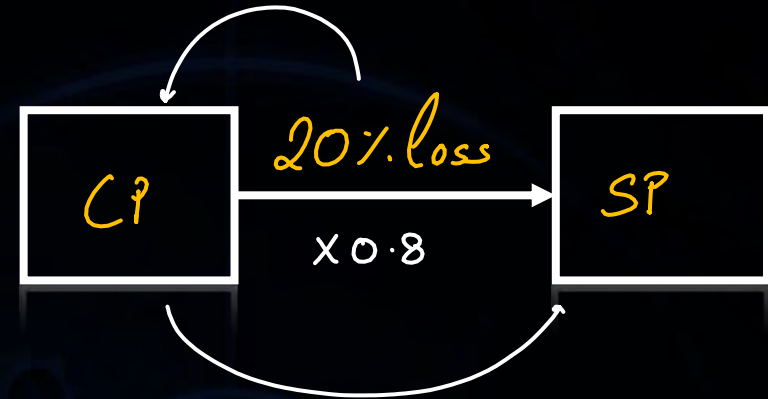
$$\text{loss}\% = \frac{\text{loss}}{CP} \times 100$$

$$CP \left[1 - \frac{\text{loss}\%}{100} \right] = SP$$



$$1.1CP = SP$$

$$\text{Profit} = 0.1CP$$



$$0.8CP = SP$$

$$\text{loss} = 0.2CP$$

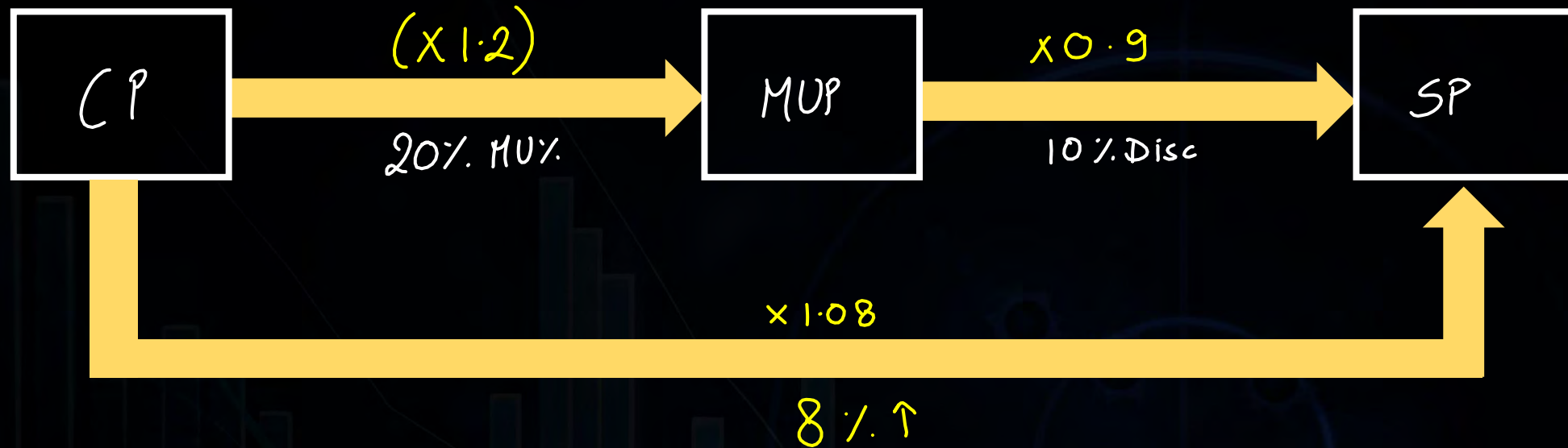


$$\text{Net \% change} = \text{MU\%} - \text{D\%} - \frac{(\text{MU\%})\text{D\%}}{100}$$

$$\text{MP} \left(1 - \frac{\text{D\%}}{100} \right) = \text{SP}$$

$$\text{CP} \left[1 + \frac{\text{MU\%}}{100} \right] = \text{MP}$$

$$\text{CP} \left[1 \pm \frac{\text{P/L\%}}{100} \right] = \text{SP}$$



$$1.2 \text{ CP} = \text{MUP}$$

$$1.08 \text{ CP} = \text{SP}$$

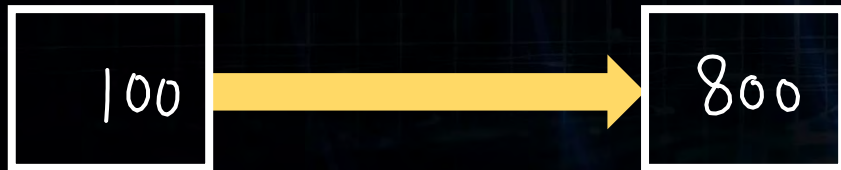
$$0.9 \text{ MUP} = \text{SP}$$

STATEMENT: A faulty balance shows 750 gm when it is actually 1000 gm.



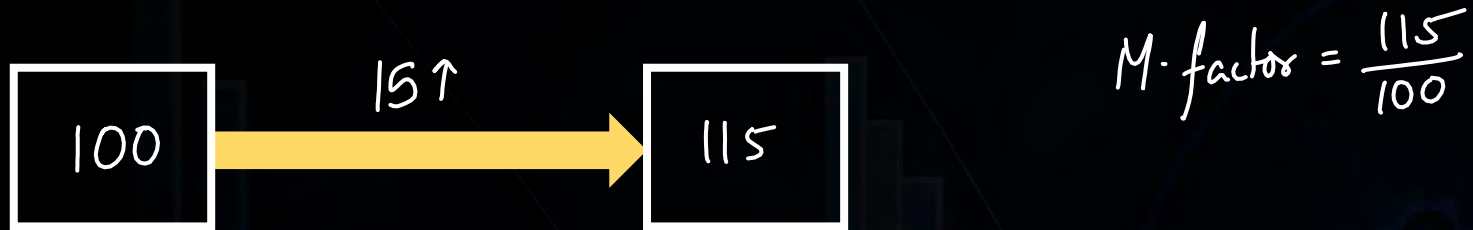
$$M.Fac = \frac{750}{1000}$$

STATEMENT: A faulty balance shows 800 gm when it is actually ~~1000~~¹⁰⁰ gm.



$$M.factor = \frac{800}{100}$$

STATEMENT: A shopkeeper mixes impurities which is 15% of the initial total paddy.



STATEMENT: A shopkeeper mixes 5 Kg impurities in 30 Kg Rice.

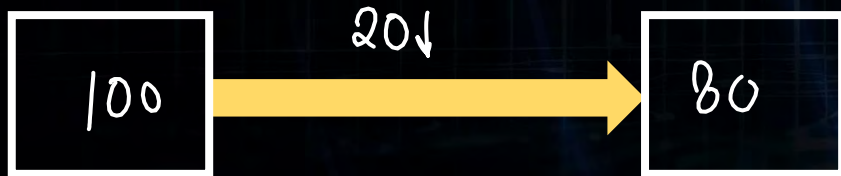


STATEMENT: A shopkeeper marked up his goods by 10%.



$$M\text{-factor} = \frac{110}{100}$$

STATEMENT: A shopkeeper gave discount of 20%



$$M\text{-factor} = \frac{80}{100}$$

$$CP = 640$$

faulty balance

$$800\text{gm} \longrightarrow 1000\text{g}$$

IMPURITY

10% impurity

$$100 \xrightarrow{10} 110$$

Disc $\rightarrow$ 25%

$$100 \xrightarrow{25\downarrow} 75$$

MD% $\rightarrow$ 10%

$$100 \longrightarrow 110$$

$$\frac{5}{4}$$

$$\frac{11}{10}$$

$$\frac{3}{4}$$

$$\frac{11}{10}$$

$$\begin{array}{ccc} CP & & SP \\ 640 & \xrightarrow{\quad} & 726 \\ & \times \frac{363}{320} & \end{array}$$

$$\begin{array}{ccc} \checkmark \text{ PROFIT} & & \\ 320 & \xrightarrow{43} & 363 \end{array}$$

$$\frac{43}{320} \times 100$$

x M factor

$$\frac{5}{4} \times \frac{11}{10} \times \frac{3}{4} \times \frac{11}{10}$$

$$\begin{array}{r} 121 \times 3 \\ \hline 1600 \\ 320 \end{array}$$

$$\frac{363}{320}$$

CHEAT SHEETS

A horizontal glowing blue pen with gold-colored tips, positioned behind the text.

SI AND CI

SIMPLE INTEREST



Interest every year = $r\%$ of P

Simple Interest = $PRT/100$

Amount = $P(1+RT/100)$

$\checkmark$
 $8\% \rightarrow \text{rate}$
 $5\text{yr} \rightarrow \text{time}$

$$SI = 40\% \text{ of } P = 0.4P$$

$$A = 1.4P$$

$$\text{Int/yr} = 0.08P$$

$$SI = 0.36P \quad A = 1.36P$$

$$\text{RATE} = 6\% \quad \text{Time } 6\text{yr}$$

Simple Interest and Series

260 280 300 320 340 360 380 400

100 lb in 5 yr

$P = \text{??}240$

$\lambda = \text{??}$

$I/\text{yr} = \text{??}20$

$\frac{1}{12} \times 100$

8.33%

6 yr $\rightarrow$ $\boxed{\text{SI } 48\% \text{ of } P}$

8% $\rightarrow$ $\boxed{\text{RATE} = 8\%}$

COMPOUND INTEREST



$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^n$$

$$\text{Compound Interest} = A - P$$

$$\text{rate} = 20\% \quad \text{Time} = 5 \frac{1}{2}$$

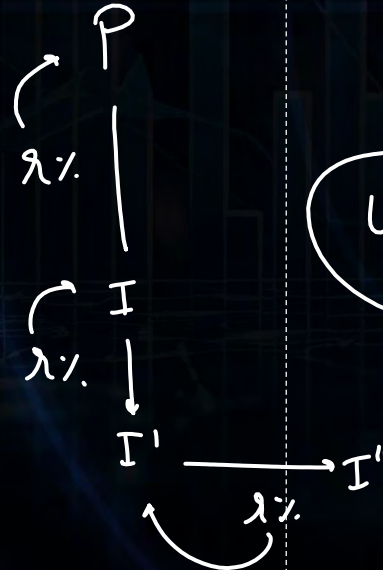
$$\text{M. factor} = 1.2$$

$$A = P[1.2]^5$$

$$CI = A - P$$

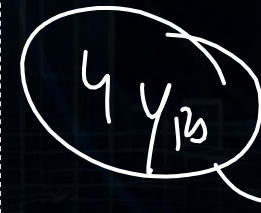
Compound Interest (2 Years)

$$CI = 2I + I'$$



Compound Interest (3 Years)

$$CI = 3I + 3I' + I''$$



$$CI = 4I + 6I' + 4I'' + I'''$$

Compound Interest

(r% per annum compounded half yearly)

$$\text{Amount} = P \left(1 + \frac{R}{200} \right)^{2n}$$

Compound Interest

(r% per annum compounded Quarterly)

$$\text{Amount} = P \left(1 + \frac{R}{400} \right)^{4n}$$

Difference in SI and CI

TIME - 2 YEARS

$$CI = 2I + I'$$

$$SI = 2I$$

$$I' = 2\% \text{ of } I$$

$$= 2\% \text{ of } 2\% \text{ of } P$$

$$I' = \frac{2^2 P}{10000}$$

TIME - 3 YEARS

$$CI = 3I + 3I' + I''$$

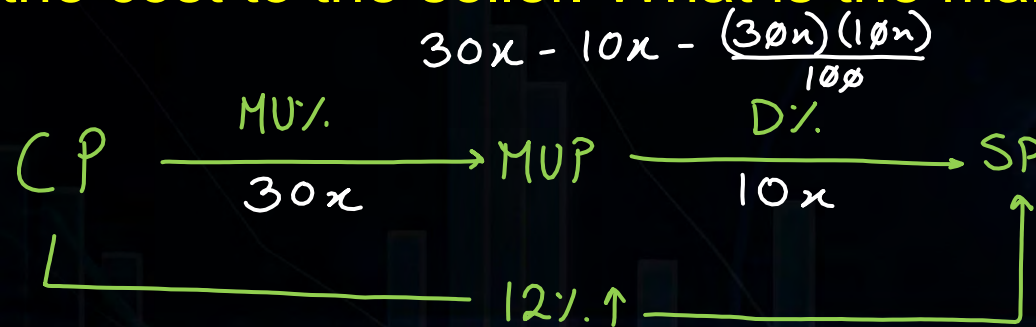
$$SI = 3I$$

$$\text{Diff} = 3I' + I''$$

$$= 3I' + 2\% \text{ of } I'$$

$$\text{Diff} = I' \left[3 + \frac{2}{100} \right]$$

A seller increases the cost price of an item by a certain percentage before offering a discount. If the markup percentage is three times the discount percentage and the resulting profit is 12%. If the discount was not provided, the total profit would have been less than the cost to the seller. What is the markup percentage?



$$CP \xrightarrow[\text{20}\%]{\frac{30x}{180\% \cdot x}} SP$$

$$x = 6 \quad x = \frac{2}{3}$$

$$20x - 3x^2 = 12$$

$$3x^2 - 20x + 12 = 0$$

Two vendors are selling bananas. [Vendor A earns a profit equivalent to the selling price of 6 bananas on selling 36 bananas.] [Vendor B, on the other hand, earns a profit equal to the cost price of 4 bananas when he sells 20 bananas]

What is the difference in their profit percentages (in percentage points)?

✓
36

SP=1 Vendor A

Profit = 6SP

30 $\xrightarrow{6}$ 36

$\frac{1}{5} \rightarrow 20\% \text{ profit}$

CP=1 Vendor B Sell 20 Ban

P = CP of 4

20 $\xrightarrow{4}$ 24

$\frac{1}{5} \rightarrow 20\%$

Shardul Thakur, a fast pace bowler in the Indian cricket team, lent a sum of Rs 80,000 in two parts. The first part is lent out at 20% per annum simple interest for two years, and the second part is lent out at 10% per annum compound interest, compounded annually for the same time period. If the total interest earned at the end of two years is Rs 26400, then what is the ratio of the sum of money lent in 1st part to that in 2nd part?

Handwritten solution:

Let the first part be x and the second part be y .

For the first part (Simple Interest):

$$SI = \frac{x \times 20 \times 2}{100} = \frac{40x}{100} = 40\%$$

For the second part (Compound Interest):

$$CI = \frac{y \times 10 \times 2}{100} = \frac{20y}{100} = 20\%$$

Total interest earned is Rs 26400, which is 33% of the total sum of Rs 80,000:

$$33\% \text{ of } 80,000 = \frac{33}{100} \times 80,000 = 26,400$$

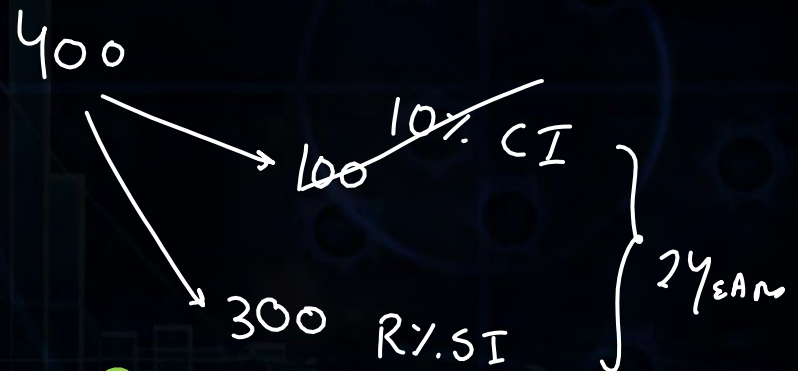
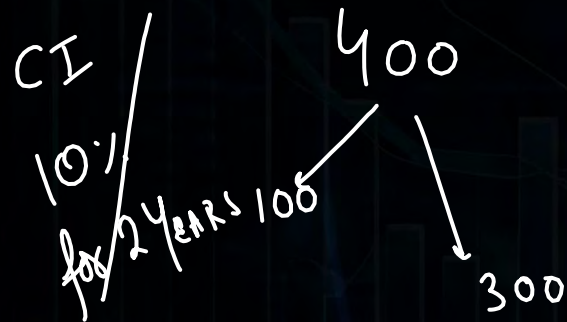
Since the total interest is 33%, and the first part contributes 40%, the second part must contribute 21% (33% - 12% = 21%).

Ratio of the sum of money lent in 1st part to that in 2nd part is:

$$12 : 7$$

Shokie Agrawal and Shanty Mukherjee, who are in the elite business of Tata Sons group, have equal amounts. Shokie invested his entire amount at 10% compounded annually for two years, and Shanty invested at $\frac{1}{4}$ at 10% compound interest (annually) and the rest at R% per annum at simple interest for the same two years period. The amount received by both at the end of 2 years is the same. What is the value of $(2R + 9)$? = $\boxed{30}$ ✓

$$A = P \left[1 + \frac{RT}{100} \right]$$



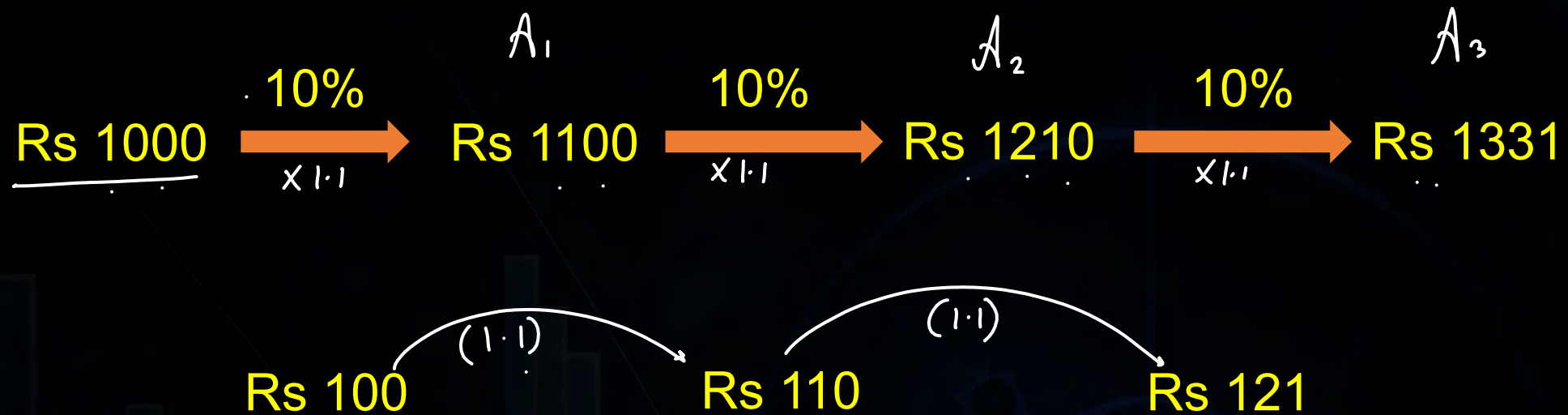
$$0.21 = \frac{2R}{100}$$

$$\boxed{21 = 2R}$$

$$300 (1.1)^2 = 300 \left[1 + \frac{2R}{100} \right]$$

$$1.21 = 1 + \frac{2R}{100}$$

CI and GP - CI in nth Year Concept



$R = 1.05$
CI in 3rd Year = 80000 → a_3

Rate % = 5%

CI in 5th year = ?

→ a_5

$$80000(1.05)(1.05)$$

CI in 3rd Year = 11025

Rate % = 5%

P = ??

$$P(1.05)(1.05)(1.05) = 11025$$

Instalments : Deposits and EMI



deposit every month

$$A = \frac{c}{i} ((1+i)^n - 1)$$

RD → Deposits $r = 10\%$

$c = 10,000/\text{year}$

$i = \frac{10}{100} = 0.1$

$n = 4\text{ yrs}$

$$A = \frac{10000}{0.1} [(1.1)^4 - 1]$$

EMI every month

$$P = \frac{c}{i} (1 - (1+i)^{-n})$$

loan

$\frac{r}{100}$

year

$$P = 200000$$

$$200000 = \frac{c}{0.1} [1 - (1.1)^{-3}]$$

An Investor invests a fixed amount every year at an annual rate of 10% compounded annually. If the total amount he has at the end of 4 years is equal to Rs 9282. What is the annual deposit he make ?

$$A = \frac{C}{i} \left[(1+i)^n - 1 \right] \quad \checkmark$$

$$9282 = \frac{C}{0.1} \left[(1.1)^4 - 1 \right]$$

$$928.2 = C [1.4641 - 1]$$

$$\frac{9282}{2000} = 0.4641 C$$

$$2000 = C$$

$$\begin{array}{r} 1.21 \\ 1.4241 \\ \quad 4 \\ 1.4641 \end{array}$$

John borrowed Rs. 2,10,000 from a bank at an interest rate of 10% per annum, compounded annually. The loan was repaid in two equal instalments, the first after one year and the second after another year. The first instalment was interest of one year plus part of the principal amount, while the second was the rest of the principal amount plus due interest thereon. Then each instalment, in Rs., is:[TITA]

$$P = \frac{c}{i} \left(1 - (1+i)^{-n} \right)$$

$$210000 = \frac{c}{0.1} \left(1 - (1.1)^{-2} \right)$$

$$21000 = c \left[1 - \frac{1}{1.21} \right]$$

$$210000 = c \left[\frac{1.21}{1.21} \right]$$

$$\boxed{121000 = c}$$

Q.

QUESTION - 7



Anil borrows Rs 2 lakhs at an interest rate of 8% per annum, compounded half-yearly. He repays Rs 10320 at the end of the first year and closes the loan by paying the outstanding amount at the end of the third year. Then, the total interest, in rupees, paid over the three years is nearest to

A) 45311

B) 51311

C) 33130

D) 40991

$r = 4\%$

14R

$$\begin{array}{r}
 P \\
 200000 \\
 \xrightarrow{\quad} I \\
 8000 \\
 \xrightarrow{\quad} I' \\
 320 \\
 2I + I' \\
 CI = \boxed{16320 \text{ Rs}} \\
 - 10320 \\
 \hline
 6000
 \end{array}$$

14R
200
34R
200
24R
4BAAR

200000	40000
33000	6000
1020	

$$(1.04)^4$$

$$(1.0816)^2$$

$$\approx (1.08)^2$$

$$\approx 1.1664$$

10,320

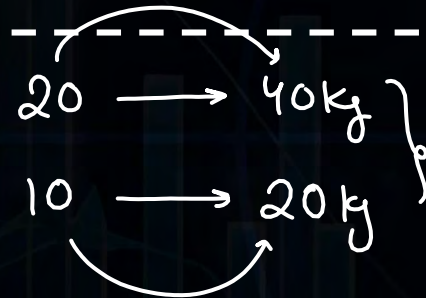
CHEAT SHEETS

AVERAGE

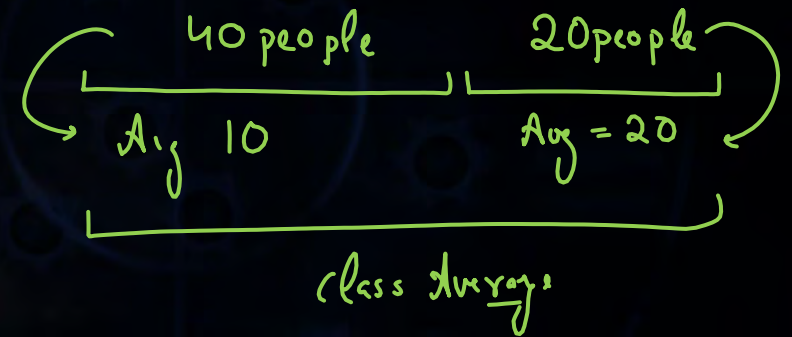
$$\text{Average} = \frac{\text{Sum}}{n}$$

$$\text{Sum} = n \times \text{Average}$$

$$\text{Weighted Average} = \frac{n_1 A_1 + n_2 A_2 + n_3 A_3}{n_1 + n_2 + n_3}$$



$$\begin{aligned} \text{Avg} &= \frac{800 + 200}{30} \\ &= \frac{1000}{3} \end{aligned}$$



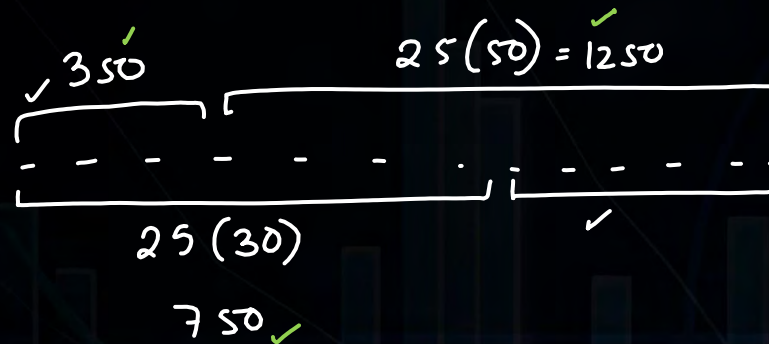
$$\frac{400 + 400}{60} = \frac{800}{60} = \frac{40}{3}$$

Q.

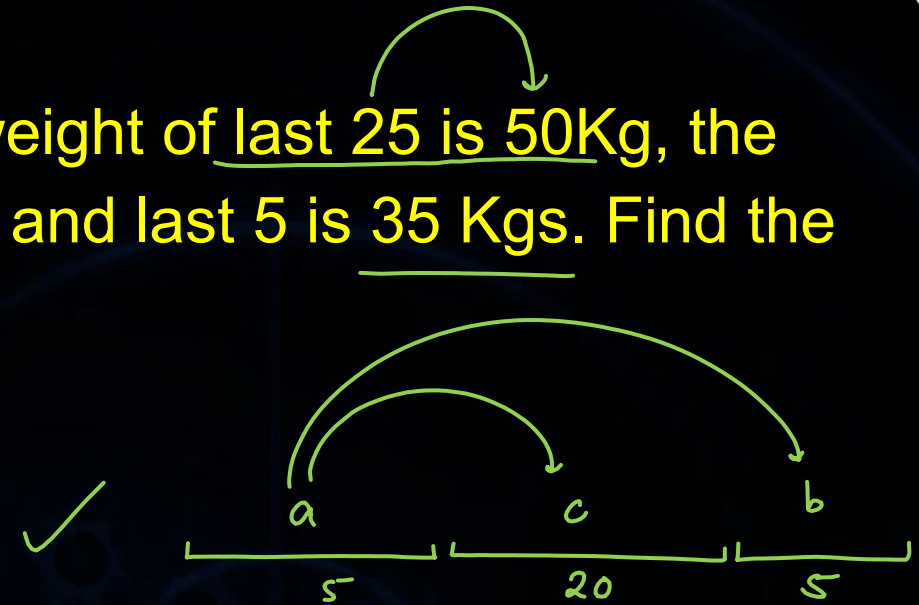
QUESTION - 8



There are 30 Students in the class, the average weight of last 25 is 50Kg, the average of first 25 is 30 Kg. The average of first 5 and last 5 is 35 Kgs. Find the average weight of the entire class.



$$Avg = \frac{2350}{60}$$



$$b + c = 1250$$

$$a + c = 750$$

$$a + b = 350$$

$$2(a + b + c) = 2350$$

Priya has 15 bags of marbles, and the average number of marbles per bag is 25. Each bag contains at least 15 marbles, and no two bags have the same number of marbles. What is the maximum possible number of marbles in any one bag

$$\begin{array}{r} \text{SUM} = 15 \times 25 \\ = 375 \end{array}$$

15 16 17 18 19 20 21 22 23 24 25 26 27 28 74 ↑

$$\frac{74}{2} (43)$$

✓ (301)

Consider six distinct natural numbers such that the average of the two smallest numbers is 10, and the average of the two largest numbers is 42. Then, the maximum possible value of the average of these six numbers is

$$\begin{array}{ccccccc}
 \underline{\quad} & < & \underline{\quad} & < & \underline{39} & < & \underline{40} & < & \underline{41} & < & \underline{43} \\
 \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & & & \underbrace{\hspace{1.5cm}} & & & & \\
 20 & & & & & & 84 & & & & \\
 & & & & \underbrace{\hspace{10cm}} & & & & & & \\
 & & & & \frac{\text{Sum}}{6} = \frac{20+84+79}{6} & & & & & &
 \end{array}$$

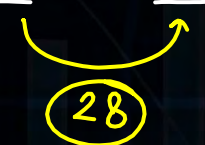
Consider six distinct natural numbers such that the average of the two smallest numbers is 10, and the average of the two largest numbers is 42. Then, the minimum possible value of the average of these six numbers is

$$\begin{array}{ccccccc}
 & 1 & 10 & 20 & 21 & & \\
 \hline
 & 9 & < & 11 & < & 12 & < & 13 & < & _ & < & _ \\
 \hline
 & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & \underbrace{\hspace{1.5cm}} & & & & & \\
 & 20 & & 25 & & 84 & & & & & & &
 \end{array}$$

$$\begin{array}{r}
 20 + 84 + 25 \\
 \hline
 6
 \end{array}$$

Consider six distinct natural numbers such that the average of the two smallest numbers is 10, and the average of the two largest numbers is 42. Then, the maximum difference between 3rd and 4th largest number is.

$$\underbrace{\underline{9} < \underline{11}}_{20} < \underline{12} < \underline{40} < \underbrace{\underline{41} < \underline{43}}_{84}$$





QUESTION - 13



The average marks of 6 students in a test is 70. All the students got distinct integral marks, one of the students obtained 80 marks and all the other students scored 50 or above. What can be the maximum possible difference between second highest and the second lowest marks.

110

50	51				
50	51				
50	51				

EXTRA SUPPORT



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Simple & Compound Interest (SI CI) One Shot | CAT 2025...
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Discount MRP and Faulty Weights | Arithmetic Ep. 10 |...
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Profit and Loss | Arithmetic Ep.09 | CAT 2025 Prep
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THANK YOU

